

Homework 6

Problem 1

For many animal species, the lengths of particular bones follow a Normal distribution. For adult male humans, the length of the femur (thigh bone, the longest bone in humans) has an approximate Normal distribution with mean 48.2 cm and standard deviation of 4.5 cm.

Complete each of the following parts by standardizing, drawing appropriate plots, and then providing an educated ballpark estimate based on the empirical rule. Then you can use software to get more precise values. But software should be your last step, not the first.

1. What percent of adult male humans have a femur length above 50cm?
2. What percent of adult male humans have a femur length between 40cm and 50cm?
3. What is the 99th percentile of femur lengths for adult male humans?

Problem 2

Daily high temperatures (degrees Fahrenheit) in San Luis Obispo in August follow (approximately) a Normal distribution with a mean of 76.9 degrees F. The temperature exceeds 100 degrees Fahrenheit on about 1.5% of August days.

1. Compute the standard deviation.
2. Suppose the mean increases by 2 degrees Fahrenheit. On what percentage of August days will the daily high temperature exceed 100 degrees Fahrenheit? (Assume the standard deviation does not change.)
3. A mean of 78.9 is 1.02 times greater than a mean of 76.9. By what (multiplicative) factor has the percentage of 100-degree days increased? What do you notice?

Problem 3

The latest series of collectible Lego Minifigures contains 3 different Minifigure prizes (labeled 1, 2, 3). Each package contains a single unknown prize. Suppose we only buy 3 packages and we consider as our sample space outcome the results of just these 3 packages (prize in package 1, prize in package 2, prize in package 3). For example, 323 (or (3, 2, 3)) represents

prize 3 in the first package, prize 2 in the second package, prize 3 in the third package. Let X be the number of distinct prizes obtained in these 3 packages. Let Y be the number of these 3 packages that contain prize 1. Suppose that each package is equally likely to contain any of the 3 prizes, regardless of the contents of other packages; let P denote the corresponding probability measure. There are 27 possible, equally likely outcomes

box1	box2	box3	X	Y
1	1	1		
2	1	1		
3	1	1		
1	2	1		
2	2	1		
3	2	1		
1	3	1		
2	3	1		
3	3	1		
1	1	2		
2	1	2		
3	1	2		
1	2	2		
2	2	2		
3	2	2		
1	3	2		
2	3	2		
3	3	2		
1	1	3		
2	1	3		
3	1	3		
1	2	3		
2	2	3		
3	2	3		
1	3	3		
2	3	3		
3	3	3		

1. Evaluate X and Y for each of the outcomes.
2. Construct a two-way table representing the joint distribution of X and Y .
3. Specify the marginal distribution of X and compute $E(X)$.
4. Specify the marginal distribution of Y and compute $E(Y)$.
5. Compute $\text{Cov}(X, Y)$
6. Find the conditional distribution of Y given $X = x$ for each possible value of x of X .
7. Compute and interpret $E(Y|X = x)$ for each possible value of x of X .

8. Find the conditional distribution of X given $Y = y$ for each possible value of y of Y .
9. Compute and interpret $E(X|Y = y)$ for each possible value of y of Y .
10. Are X and Y independent? Explain; support your answer with appropriate numbers.

Problem 4

Xavier and Yolanda are playing roulette. They both place bets on red on the same 3 spins of the roulette wheel before Xavier has to leave. (Remember, the probability that any bet on red on a single spin wins is $18/38$.) After Xavier leaves, Yolanda places bets on red on 2 more spins of the wheel. Let X be the number of bets that Xavier wins (out of 3) and let Y be the number that Yolanda wins (out of 5).

1. Identify by name the marginal distribution of X . Be sure to specify the values of any relevant parameters. Compute $E(X)$.
2. Identify by name the marginal distribution of Y . Be sure to specify the values of any relevant parameters. Compute $E(Y)$.
3. The joint distribution of X and Y is represented in the table below. Explain why $p_{X,Y}(1, 4) = 0$ and $p_{X,Y}(2, 1) = 0$.
4. Compute $p_{X,Y}(2, 3)$. (Yes, the table tells you it's 0.1766, but you have to show how this number can be computed based on the assumptions of the problem.)
5. Compute $\text{Cov}(X, Y)$.
6. Are X and Y independent? Explain; support your answer with appropriate numbers.

x, y	0	1	2	3	4	5
0	0.0404	0.0727	0.0327	0	0	0
1	0	0.1090	0.1963	0.0883	0	0
2	0	0	0.0981	0.1766	0.0795	0
3	0	0	0	0.0294	0.0530	0.0238